

PROPOSAL TO THE CPO & GMS OF PRODUCT

Measuring What Matters

A value-realization framework for product adoption in a
consumption-based security business

Principal PM, Product Adoption Analytics

Built end-to-end with a spec-driven, AI-first development workflow

CVRS

The Technical PM operating model

Specs are the contract

Markdown specs define the metric, data model, and pipeline first; code and tests are generated from them and kept in sync

Ship a prototype, not slides

Everything here runs: real BigQuery, a layered SQL pipeline, 28 automated tests, and a live Streamlit dashboard

Measure like an engineer

Every claim is reproducible and test-backed; the metric is evaluated the way a model would be, against labeled ground truth

Own the whole seam

Bridge Product, CS, and Finance - define the number, defend the trade-offs, and stand behind the implementation

Raw usage is lying to us

The task: score value realization and classify every account into an actionable health tier - at the customer, SKU, and feature level.

10%

pure shelfware - paying, zero consumption, invisible until renewal

5%

burn ~90% of entitlements in 3 months, then go dark - today scored as heroes

15%

run at 120%+ every month - expansion revenue nobody systematically works

Why leadership cares

- Renewal surprises: churn risk hides behind cumulative usage charts
- Wrong incentives: teams paid on volume they can game
- SKUs bundle many features - we measure the bundle, not whether value inside it is realized

One score, four honest components

40%

Consumption utilization

Consumed vs entitled, capped at 100% - burning faster is not healthier

30%

Feature adoption depth

Features used vs eligible, per SKU - catches shallow deployments

20%

Consistency

Trailing 3-month trend vs peak - punishes spike-and-drop within a month

10%

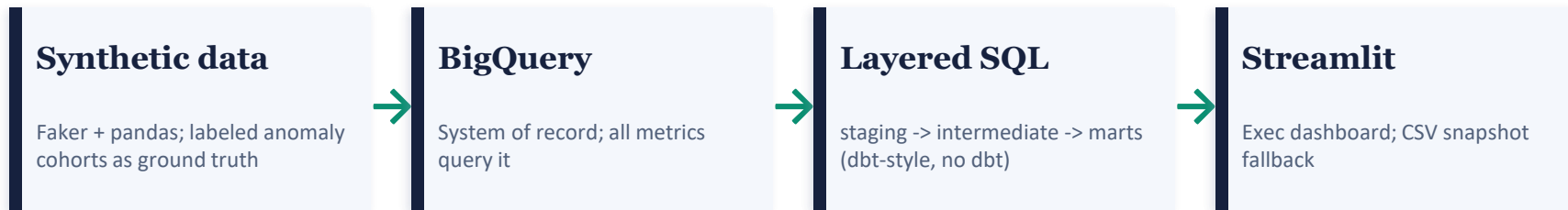
Breadth

SKUs with real usage vs owned - no credit for ignored products

Health tiers drive playbooks: **Healthy 70+** **Watch 40-69** **At Risk <40 or drop** **Shelfware = zero use**

Revenue overlay: committed ARR, ARR-at-risk, and expansion pipeline sit alongside the score - never inside it (utilization is capped; a test proves CVRS depends only on its four components).

From synthetic data to executive dashboard



Build vs Buy

Buy the warehouse (BigQuery); build the transforms as plain layered SQL - dbt's discipline without its dependency for a prototype

Latency vs Accuracy

Monthly batch grain - entitlements are monthly and QBR/renewal decisions are monthly; streaming buys nothing here

The hard part solved in SQL

Mid-year expansions summed per month via a date-range join - overlapping contracts raise the denominator, no double-counting

Designed for messy enterprise data

Spike & drop

Consistency compares this month to the trailing 3-month peak - a cliff to zero collapses the score within a month

Shelfware

Reporting is built from entitlements, not usage - zero-consumption accounts can never silently vanish

Consistent coverage

Utilization capped in the score; raw overage feeds the expansion pipeline instead of inflating health

Mid-year expansion

Denominator steps up in the expansion month; a 2-month grace window stops the expected dip reading as churn

Deployment lag

A real per-product adoption date drives day-level time-to-first-value (0-90 days), not a monthly approximation

We evaluated the metric like a model

Ground-truth validation

- Synthetic data ships with labeled cohorts - the answer key is known before scoring
- Tests require $\geq 90\%$ of each injected cohort to land in the intended health tier
- 28 automated tests: structural integrity, score bounds, no-account-dropped, and cohort detection - green on every run
- BigQuery caught real bugs a local run hid: a CTE/column name collision and negative time-to-value

Tied to business KPIs

Shelfware tier

Deployment rescue -> time-to-first-value

At Risk tier

Save motions -> gross retention

Expansion flag

True-up pipeline -> net revenue retention

Feature depth

Activation campaigns -> roadmap bets

Hard truths, and the road ahead

Hard truths learned

- The ≥ 25 -point drop rule is dormant on this data - every real decline already breaches the absolute floor first; it earns its place only for slow erosion the data doesn't model yet
- Day-precision on monthly data is a trap: TTFV needed a real adoption date, not arithmetic on a daily start
- Composite weights are judgment - so they are versioned, tested, and governance-owned, not tuned quietly

If rebuilt with today's stack

- Agentic triage: an agent investigates every tier change and drafts the CSM brief
- LLM-generated account narratives on top of the component data - the 'why', not just the score
- Keep the spec-driven loop: metric changes ship as spec diffs, with code and tests regenerated from them

The ask: endorse CVRS + health tiers, fund productionization, charter the metric governance forum. *Live demo next.*